

AI INTERVIEW COACH

Software Requirements Specification

Document Version: 1.0

Date: March 4, 2026

Status: Final Release

Professional Interview Preparation Platform

TABLE OF CONTENTS

1. Executive Summary
2. Project Scope & Objectives
3. Functional Requirements
4. System Architecture
5. Database Design
6. API Specification
7. User Interface
8. Non-Functional Requirements
9. Use Cases & Workflows
10. Security & Compliance
11. Deployment & Maintenance
12. Testing Requirements
13. Glossary & Technical Terms
14. Appendices

AI INTERVIEW COACH - SOFTWARE REQUIREMENTS SPECIFICATION

1. EXECUTIVE SUMMARY

Project: AI Interview Coach

Objective: Intelligent AI-powered interview preparation platform providing personalized coaching and realistic practice questions.

Key Features:

- User authentication & profile management
- AI-driven question generation (Mistral 7B)
- Real-time interview simulation
- Intelligent feedback analysis
- Performance tracking & analytics

Success Criteria: 95%+ uptime, sub-2s response time, 100+ active users by Q2

2. PROJECT SCOPE & OBJECTIVES

In Scope:

- User authentication (JWT-based)
- Interview session management
- AI question generation with context
- Answer submission & evaluation
- Automated feedback generation
- Performance analytics dashboard

Out of Scope:

- Video/audio recording
- Mobile app development
- Payment processing
- Third-party integrations

Target Users:

- Job seekers (18-45 age group)
- Interview practitioners
- Career switchers
- Students

3. FUNCTIONAL REQUIREMENTS

FR-1: User Management

- User registration with email verification
- Password hashing (PBKDF2)
- JWT token-based authentication
- Profile management (skills, experience, goals)
- Session timeout after 7 days

FR-2: Interview Session Creation

- Create sessions with job field, experience level, company
- Session templates (behavioral, technical, HR)
- Customizable session duration (15-60 minutes)

FR-3: Question Generation

- Generate 5-15 questions per session via Mistral API
- Questions contextualized by field/level/company
- Mix of behavioral, technical, situational questions
- Difficulty levels: Beginner, Intermediate, Advanced

FR-4: Interview Experience

- Display one question at a time
- Text input for candidate answers
- Answer word count: 50-500 words
- Session timer with warnings

FR-5: Feedback System

- AI-generated feedback on answers
- Scoring: 1-10 scale
- Improvement suggestions
- Communication analysis

FR-6: Analytics Dashboard

- View interview history
- Performance trends (charts/graphs)
- Score distribution
- Time-to-completion metrics
- Strengths & weaknesses analysis

4. SYSTEM ARCHITECTURE

Frontend Stack:

- HTML5, CSS3, JavaScript (Vanilla)

Responsive design
Real-time UI updates

Backend Stack:

Python Flask 2.x
SQLAlchemy ORM
Flask-CORS
Flask-JWT-Extended
Werkzeug password hashing

Database:

SQLite (sqlite:///interview_coach.db)
5 core tables: users, interviews, questions, answers, feedback

AI Integration:

Mistral 7B Instruct v0.2
Local API (port 1234)
OpenAI-compatible interface

Deployment:

Heroku/Vercel Ready
RotatingFileHandler logging
Environment-based configuration

5. DATABASE DESIGN

Users Table:

id (PRIMARY KEY)
email (UNIQUE)
password_hash (PBKDF2)
created_at, updated_at

Interviews Table:

id (PRIMARY KEY)
user_id (FK)
field, level, company
created_at, status

Questions Table:

id (PRIMARY KEY)
interview_id (FK)
question_text, difficulty
created_at

Answers Table:

id (PRIMARY KEY)
question_id (FK)
answer_text, submission_time
created_at

Feedback Table:

id (PRIMARY KEY)
answer_id (FK)
score (1-10), feedback_text
improvement_suggestions
created_at

6. API SPECIFICATION

Authentication Endpoints:

POST /api/auth/register - User registration
POST /api/auth/login - User login
POST /api/auth/logout - User logout

Interview Endpoints:

POST /api/interviews - Create interview session
GET /api/interviews - List user interviews
GET /api/interviews/{id} - Get interview details
GET /api/interviews/{id}/questions - Get interview questions

Question/Answer Endpoints:

POST /api/answers - Submit answer
GET /api/answers/{id} - Get answer with feedback
GET /api/interviews/{id}/feedback - Get session feedback

Analytics Endpoints:

GET /api/analytics/dashboard - User statistics
GET /api/analytics/history - Performance history
GET /api/analytics/scores - Score distribution

7. USER INTERFACE SPECIFICATIONS

Page 1 - Landing Page:

Hero section with value proposition
Feature highlights (3-4 key benefits)
Call-to-action buttons

Testimonials section

Page 2 - Login/Signup:

Email/password input forms

Form validation (real-time)

"Remember me" option

Password recovery link

Page 3 - Dashboard:

Welcome message with user name

Quick stats (total interviews, average score)

Recent interview history (table)

Start new interview button

Page 4 - Interview Setup:

Job field selector (dropdown)

Experience level selector

Company name input

Session duration selector

Start interview button

Page 5 - Interview Session:

Question display area

Timer with countdown

Answer input (textarea)

Submit answer button

Progress indicator (Q# of #)

8. NON-FUNCTIONAL REQUIREMENTS

Performance:

API response time: < 2 seconds

Page load time: < 3 seconds

Database query optimization (indexes on FK)

Caching strategy for static assets

Security:

HTTPS/TLS encryption

CSRF protection

Input validation & sanitization

SQL injection prevention (parameterized queries)

API rate limiting (100 req/min per user)

JWT expiration (7 days)

Reliability:

- 95%+ uptime SLA
- Error logging (RotatingFileHandler)
- Graceful error handling
- Database backup strategy

Scalability:

- Horizontal scaling capability
- Stateless backend design
- Database connection pooling
- CDN for static assets

Usability:

- Mobile-responsive design
- Accessibility (WCAG 2.1 AA)
- Intuitive navigation
- Clear error messages

9. USE CASES & WORKFLOWS

Use Case 1: New User Registration

1. User fills registration form
2. System validates email uniqueness
3. Password hashed & stored
4. Confirmation email sent
5. User redirected to login

Use Case 2: Interview Session Flow

1. User creates session (field, level, company)
2. Mistral API generates 10 questions
3. System displays first question
4. User enters answer
5. Upon completion, AI generates feedback
6. Session saved to database
7. Analytics updated

Use Case 3: View Performance Analytics

1. User clicks "Dashboard" menu
2. System fetches user interview history
3. Displays aggregated statistics
4. Shows score trends (chart)

5. Highlights strengths & improvement areas

10. SECURITY & COMPLIANCE

Authentication:

- JWT tokens (7-day expiration)
- Refresh token mechanism
- Password requirements (min 8 chars, mixed case, numbers)

Data Protection:

- PBKDF2 password hashing
- Database encryption at rest
- HTTPS in transit
- Principle of least privilege

Access Control:

- Users can only view own data
- Admin endpoints protected
- Role-based access control (future)

Compliance:

- GDPR-ready (data export, deletion)
- Privacy policy & ToS
- User data retention policy (1 year)

11. DEPLOYMENT & MAINTENANCE

Deployment:

- Git-based deployment (GitHub/Heroku)
- Environment variables for secrets
- Database migrations (Alembic-ready)
- Automated testing in CI/CD

Monitoring:

- Application performance monitoring (APM)
- Error tracking (Sentry integration)
- Uptime monitoring
- Database query performance

Maintenance:

- Regular security patches
- Database optimization
- Log rotation & cleanup

User support & issue tracking

12. TESTING REQUIREMENTS

Unit Tests:

- Authentication logic
- API endpoint validation
- Database model tests
- Utility function tests

Integration Tests:

- End-to-end interview flow
- Database CRUD operations
- API integration with Mistral
- JWT token validation

System Tests:

- Load testing (100+ concurrent users)
- Performance benchmarking
- Security testing (OWASP Top 10)
- Accessibility testing

13. GLOSSARY & TECHNICAL TERMS

- JWT:** JSON Web Token for authentication
- PBKDF2:** Password-based key derivation function
- ORM:** Object-Relational Mapping (SQLAlchemy)
- API:** Application Programming Interface
- CORS:** Cross-Origin Resource Sharing
- SLA:** Service Level Agreement
- WCAG:** Web Content Accessibility Guidelines

14. APPENDICES

A. Technology Stack Summary

- Python 3.8+, Flask 2.x, SQLAlchemy
- SQLite3, JWT, Werkzeug
- Mistral 7B AI Model
- HTML5/CSS3/JavaScript

B. File Structure

C. API Response Examples

Authentication Success:

Question Generation:

Feedback Response:

Document End

This SRS document contains all essential specifications for the AI Interview Coach platform. For questions, contact the development team.